Fuzzy-Calibrated Conformal Prediction for Adaptive Uncertainty-Aware AI

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Problem

Point predictions can be overconfident under noisy or shifted observations.

Method

Bootstrap ensemble disagreement $\rightarrow$ fuzzy uncertainty regime $\rightarrow$ conformal interval.

Seeded proxy result

Accuracy 0.9737; Brier 0.0210; ECE 0.0226; coverage 0.9386; interval width 0.1759.

Scope

The dataset is a methodological proxy, not clinical validation. Next: temporal physiological benchmarks and repeated seeds.